

Problem-Solution Fit

Date	9 July 2026
Team ID	Not Assigned
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] • Beginner farmers • Agricultural students • Small-scale farmers	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] • Limited technical knowledge • Lack of access to smart tools • Budget constraints	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Existing solutions: • Traditional farming advice • Local experts Pros: • Easy to access • Based on experience Cons: • Not data-driven • Inaccurate sometimes • No real-time analysis
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Difficulty in selecting suitable crops (very frequent) • Lack of knowledge about soil nutrients (frequent) • Unpredictable weather conditions (frequent)	9. PROBLEM ROOT / CAUSE [RC] • Lack of data-driven tools • Poor awareness of soil & weather analysis • No easy access to ML-based recommendations	7. BEHAVIOR + ITS INTENSITY [BE] • Depends on traditional methods (high) • Takes decisions based on guesswork (high) • Seeks advice from others (medium)

3. TRIGGERS TO ACT [TR]

- Crop failure or low yield
- Financial loss
- Need to improve productivity

4. EMOTIONS BEFORE / AFTER [EM]

Before:

- Confused
- Worried
- Uncertain

After (using OptiCrop):

- Confident
- Relieved
- Hopeful

10. YOUR SOLUTION [SL]

- OptiCrop Smart Agriculture Engine
- Uses ML model to recommend best crop
- Takes NPK, temperature, humidity, pH, rainfall as input
- Provides accurate, fast, and user-friendly recommendations

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

- Mobile apps
- Websites
- YouTube / Google

OFFLINE

- Local farmers
- Agricultural officers
- Farming experience